

BDCL

SRN: PES1PG25GA04L1

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subject code: UQ2SCA6S3A

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-> AER A civil aviation authority maintains a database to manage aircrafts, pilots and flight operation.

collections and fields.

1. aircrafts

aircraftid (string)

model (string)

manufacturer (string)

capacity (number)

status (string)

2. pilots

pilotid (string)

name (string)

license no (string)

experience years (number)

assigned Aircraft (string)

3] Flights

flightcode (string)
 source (string)
 destination (string)
 departuretime (string)
 aircraftid (string)
 Pilotid (string)

- 1 insert a new aircraft with ID A320, model A320neo manufacture Airbus, capacity 180, and status Active.

db.aircrafts.insertOne({aircrafts: "A320",
 model: "A320neo", manufacture: "Airbus",
 capacity: 180, status: "Active"})

- 2 insert two pilot records with different experience levels and assigned aircraft.

db.Pilots.insertMany([

Pilotid: "P101",
 name: "Rahul Sharma",
 licenseno: "LIC98765",
 experienceyear: 8,
 assignedAircraft: "A320"

,

{ Pilotid: "P102"

name: "Anitha Verma",

licenseno: "LIC12345",

experienceyear: 5,

assignedAircraft: "A320" })

Teacher's Signature:

3 Add three flight records operating between different cities

db. flight. insertMany (CS

flightcode: "FL001",

source: "Bangalore",

destination: "Delhi",

departureTime: "08:30",

aircraftid: "A320",

pilotid: "P101"

},

{

flightcode: "FL002",

source: "Mumbai",

destination: "Chennai",

departureTime: "12:00",

aircraftid: "B737",

Pilotid: "P101"

},

{

flightcode: "FL003",

source: "Bangalore",

destination: "Hyderabad",

departureTime: "10:15",

aircraftid: "A320"

Pilotid: "P101"

},

])

4 Display all aircraft records available in the database

-> db.aircraft.find()

5 Retrieve all pilots who have more than 5 years of flying experience

db.pilots.find({experienceyear: { \$gt: 5 } })

6 Find all flights departing from Bangalore

-> db.flights.find({source: "Bangalore"})

7 Display details of an aircraft whose status is maintenance

~~db~~-> db.aircraft.find({status: "maintenance"})

8 ~~update~~ update the status of aircraft A320 to maintenance

-> db.aircraft.updateOne({ \$set: { status: "maintenance" }, { aircraftid: "A320" })

9 Change the assigned aircraft of a pilot using their pilotid

-> db.pilots.updateOne({ assignedaircraft: "B738" }, { \$set: { } })

-> db.pilots.updateOne({pilotId: "P103"}, { \$set: {assignedaircraft: "B708"} })

10 modify the departure time of a specific flight

-> db.flights.updateOne({flightCode: "FL003"}, { \$set: {departuretime: '8:00'} })

11 Delete a pilot record using pilotId

-> db.pilots.deleteOne({pilotId: "P103"})

12 Remove all flights whose destination is Chennai

-> db.flights.deleteMany({destination: "Chennai"})

13 Delete an aircraft record that is no longer in service.

-> db.aircraft.deleteOne({aircraftid: 'A320'})